

The Four-Form Determinant Table and Its Unique Positive-Shift Degeneracy

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Abstract

The Fejer shift produces six pairwise determinants 2 , s -squared times h , $2+qh$, $qh-2$, a -squared times h , and 2 . For positive h and odd $q=as$, the only zero is $q=1, h=2$, where the middle two forms coincide. The classification is `ALGEBRAIC_REFINEMENT_L1` and the verdict is `PROVED_UNIQUE_DEGENERACY_Q1_H2`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains STOP-SCOPED [2].

2 Four shifted forms

The Fejér/four-sign reduction itself is imported from TPC-130 [1]. The new L1 content below is the complete six-entry cross-determinant table and its degeneracy classification. Let $a, s \in \mathbb{Z}_{>0}$ be the positive affine slopes and put $q = as$. Then

$$\begin{aligned} L_1 &= d + sz, & L_2 &= u + az, \\ L_3 &= d + s(z + h), & L_4 &= u + a(z + h). \end{aligned}$$

For $L_i = b_i + m_i z$, use $\det(L_i, L_j) = m_i b_j - m_j b_i$.

Theorem 1 (Complete determinant table). *If $a, s > 0$ and $su - ad = 2$, then*

(i, j)	$(1, 2)$	$(1, 3)$	$(1, 4)$	$(2, 3)$	$(2, 4)$	$(3, 4)$
$\det(L_i, L_j)$	2	$s^2 h$	$2 + qh$	$qh - 2$	$a^2 h$	2

For positive integer h and positive odd integer q , the unique vanishing entry is $\det(L_2, L_3) = 0$ at $q = 1, h = 2$. At that point $a = s = 1$, $u - d = 2$, and $L_2 = L_3$.

Proof. Expansion gives the six entries. All except $qh - 2$ are visibly nonzero. The equation $qh = 2$ with q positive odd and h positive integral forces $q = 1, h = 2$. Then $as = 1$ forces $a = s = 1$, and $su - ad = u - d = 2$, proving coincidence. $\square$

3 Refined arithmetic gate

The exceptional shift is algebraically simpler because $\mu(L_2)\mu(L_3) = \mu(L_2)^2$. Every other positive shift is a nondegenerate four-form Möbius correlation. This table replaces the vague phrase “four-sign barrier” by one exceptional cell plus a precise growing nondegenerate family.

4 Loss, level and scope ledger

The smallest missing literal theorem is

NONDEGENERATE_GROWING_FOUR_MOBIUS_CORRELATION_BOUND.

The next route is FEJER_NONDEGENERATE_SHIFT_ROUTE. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

5 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Liang Wang. Fejer four-sign h3 gate, 2026.
- [2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.